Riya Palesha

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Education

University of Massachusetts Amherst **2026 – 2028**
MSc Computer Science *GPA : 4.0*

Pune Institute of Computer Technology **2020 – 2024**
B.E. Computer Engineering *GPA : 9.53*

Indian Institute of Technology, Madras **2021 – 2026**
BSc Programming and Data Science *GPA : 8.70*

Technical Skills

Programming Languages : Python, C++ JavaScript, SQL, HTML/CSS
Backend & Web Technologies : REST APIs, Node.js, FastAPI, Flask, React, Next.js, Vue.js, Express, PyData Stack
Cloud & DevOps : AWS, GitLab CI/CD, Jenkins, Grafana, Prometheus
Databases & Storage : PostgreSQL, MongoDB, Aerospike
Core Concepts : Object-Oriented Programming (OOP), Data Structures & Algorithms, Artificial Intelligence, Machine Learning
Developer Tools : Git, GitHub, Jira, Postman, VS Code, Eclipse, AI-Assisted Engineering

Work Experience

Barclays | *Software Engineer - BA4* **July 2024 – Aug 2026**

- Optimized system observability across distributed microservices, **reducing mean time to detect (MTTD) operational incidents by 45%**, by implementing real-time Grafana monitoring dashboards and Prometheus alert pipelines.
- Architected and deployed highly available, multi-region AWS disaster recovery solutions for the Fraud Platform, ensuring zero downtime during regional outages through modular CloudFormation templates.
- Managed complex quarterly database restacking operations across **51 EC2 nodes**, ensuring flawless execution with **zero data loss or service downtime** across environments.
- Automated Aerospike cluster configuration workflows, **cutting deployment errors by over 70%** and eliminating manual intervention, by building GitLab CI/CD pipelines and infrastructure-as-code scripts.
- Streamlined infrastructure provisioning across development and production environments, **reducing setup latency by 80%**, by developing modular CloudFormation Templates, AWS Lambda functions and Gitlab pipelines.
- Spearheaded infrastructure setup and IAM access controls for an AWS MSK (Kafka) cluster, configuring scoped roles and custom alerts that enabled **5+ application teams** to securely deploy operations.
- Eliminated hardcoded database credentials by architecting the aerospike-secret-agent service, automating secret retrieval from a central vault to AWS and injecting credentials securely into database configurations via Chef recipes.

Barclays | *Summer Intern* **June 2023 – Aug 2023**

- Automated enterprise Jira governance workflows, **cutting manual project tracking effort by over 60%**, by scripting Groovy automation tools integrated with Jenkins CI/CD pipelines.
- Designed responsive UI components for the Fraud Platform Dashboard using React, enhancing usability and operational visibility for internal teams.

Projects

ASKIVA: AI-Powered Academic Chatbot | *Next.js, Tailwind CSS, Flask, SQLAlchemy, ChromaDB, LangChain* **March 2025**

- Architected ASKIVA as a full-stack, AI-powered web application using Next.js and Flask, delivering an intuitive interface that automated query resolution and improved student concept comprehension.
- Engineered a secure SQLAlchemy backend with JWT authentication and role-based dashboards, empowering administrators to seamlessly manage content and monitor user engagement.

Data-Driven Business Performance Analysis | *Python, EDA, Data Visualization, Business Analytics* **October 2024**

- Conducted exploratory data analysis across **2.5+ years of financial and inventory records**, evaluating transactional ledgers for **10 clients** and inventory movement across **7 product categories** to identify high-demand SKUs and cash flow leakages.
- Delivered actionable inventory and payment restructuring strategies that optimized stock turnover ratios and addressed key service delivery pain points identified through customer sentiment feedback.

Reducing Hallucinations through Enhanced Retrieval-Augmented Generation Pipelines **Jan 2026**

- Engineered a document Q&A evaluation framework across **250+ benchmark queries** using Ollama, LangChain, and confidence scoring, **reducing LLM hallucination rates to <2%**.
- **Achieved 94% factual retrieval precision** across multi-format documents (PDF, CSV, DOCX) by implementing semantic chunking, hierarchical partitioning, query reformulation, reranking, and FAISS/ChromaDB vector indexing.

Leadership / Extracurricular

- Leveraged AI coding assistants to rapidly brainstorm and prototype GenAI-based solutions during **Barclays Hackathons**, accelerating the delivery of compliant CloudFormation Templates and fraud-detection rules.
- Awarded **Best Capstone Project** for developing a Library Management System at IIT Madras.
- Rotaract Club of Pune Phoenix: Contributed to community service initiatives focused on health and youth development.
- PICT CSI: Organized technical **workshops for 100+** students, fostering peer learning and collaboration.